Exercise Sheet 12

The Grimoire Closes · two sessions

House rules for the project.

- First: `pip install -r requirements.txt`. Then create **two** files in `my_work/`: `paygap.py` (your functions; it never prints and never reads a file by itself) and, per exercise, a small `checkN.py` that imports it and prints. In session 2 you add `report.py`.
- Run everything from the lesson folder: `python my_work/check1.py`. Because the check script lives next to `paygap.py`, `import paygap` just works.
- Every function gets a one-line docstring. Future Jen will thank present Jen.
- Hand in at the end of **each** session: commit, push to `lesson-12`. Open the pull request after session 2.

The rules of the game: the Accord of Equal Coin *read first*

The Accord requires six figures, the same six the UK regulations ask of every employer with 250+ staff. The Treasury roll has 200 names, so in the real world it would be voluntary, which is exactly why the Queen wants it done properly. All gaps are a percentage of **men's** pay:

```
gap % = (men - women) / men * 100          # positive: men earn more
```

FIGURE	BASED ON
1. Mean hourly pay gap	mean hourly rate of men vs women
2. Median hourly pay gap	median hourly rate of men vs women
3. Mean bonus gap	mean bonus, counting only people who got one
4. Median bonus gap	median bonus, same people
5. Proportion receiving a bonus	% of all men and % of all women who got any bonus
6. Pay quartiles	sort everyone by hourly rate, cut into four equal bands, % women in each

Hourly rate = `annual_salary / 52 / weekly_hours`. That's the whole reason part-timers don't skew the result.

Session 1 · the numbers

Exercise 1 Load and add the hourly rate *15 min · `paygap.py`, `check1.py`*

> **RAVEN #1201 from Mistress Pell:** *"Start with the only number that matters: what does each person earn per hour? Sabra at the Harbour works 18.75 hours; her salary looks tiny and her hourly rate is fine. The Grimoire never understood that."*

GIVEN (TOP OF PAYGAP.PY)

```
"""paygap.py: the Accord of Equal Coin, by the book."""
import pandas as pd
```

(a) Write `load_payroll(path)`: read the CSV, add a column `hourly_rate`, return the DataFrame.

(b) In `check1.py`: `import paygap`, load `data/thornmere_payroll.csv`, print `df.shape`, the first three rows of `employee_id, gender, weekly_hours, annual_salary, hourly_rate` rounded to 2 decimals, and the mean hourly rate of women and of men.

EXPECTED OUTPUT

```
(200, 13)
  employee_id gender  weekly_hours  annual_salary  hourly_rate
0      TM1092      M          37.5          41900        21.49
1      TM1178      F          37.5          26550        13.62
2      TM1105      F          37.5          24950        12.79
Mean hourly rate, women: 14.95 cr
Mean hourly rate, men:   18.62 cr
```

Exercise 2 The hourly pay gap

15 min · `check2.py`

> **RAVEN #1202 from Queen Aldara:** *"Two numbers. Mean and median. We are told they differ, and that the difference 'tells a story'. We would like the story."*

(a) In `paygap.py`, write `gap_percent(men, women)`: the formula from the rules, rounded to 1 decimal, returned as a plain `float` (wrap it: `float(round(..., 1))`).

(b) Write `hourly_gap(df)`: pick men's and women's hourly rates and return a dict with keys `mean_gap` and `median_gap`, both made with `gap_percent`.

(c) `check2.py` prints `gap_percent(20.00, 17.40)`, `gap_percent(18.50, 18.90)` and `hourly_gap(df)`.

EXPECTED OUTPUT

```
13.0
-2.2
{'mean_gap': 19.7, 'median_gap': 18.7}
```

Hint: Why both? The mean is pulled up by a few very high salaries (mostly men). The median is the "typical" person. If the mean gap is much bigger than the median gap, the problem is at the top.

Exercise 3 The bonus gap

15 min · `check3.py`

> **RAVEN #1203 from Lady Siv of the Rewards:** *"Bonuses are my office and I already know I won't like this. Only count people who actually got one. A zero isn't a small bonus, it's no bonus."*

(a) Write `bonus_gap(df)`. Keep only rows with `bonus > 0` for the gaps. Return a dict with `mean_bonus_gap`, `median_bonus_gap`, `pct_men_with_bonus` and `pct_women_with_bonus` (the last two out of *all* men / *all* women, 1 decimal).

(b) `check3.py` loops over the dict with `.items()` and prints `key: value` per line.

EXPECTED OUTPUT

```
mean_bonus_gap: 22.8
median_bonus_gap: 30.8
pct_men_with_bonus: 66.7
pct_women_with_bonus: 45.8
```

Exercise 4 Pay quartiles

15 min · `check4.py`

> **RAVEN #1204 from Mistress Pell:** *"The quartiles are the picture the Privy Council will remember. Who's in the bottom 25%? Who's in the top?"*

(a) Write `pay_quartiles(df)`: `pd.qcut(df["hourly_rate"], 4, labels=[...])` with the labels `Lower, Lower middle, Upper middle, Upper`; then `pd.crosstab(bands, df["gender"], normalize="index") * 100`, rounded to 1 decimal.

(b) Rename the columns to `pct_women` and `pct_men`, name the index `quartile` (`table.index.name = "quartile"`), and return the columns in that order.

(c) `check4.py` prints it.

EXPECTED OUTPUT

```
gender        pct_women  pct_men
quartile
Lower              72.0     28.0
Lower middle       46.0     54.0
Upper middle       30.0     70.0
Upper              18.0     82.0
```

Hint: Read it like a sentence: "72% of the lowest-paid quarter are women; 18% of the highest-paid quarter are women." Then look at the headcount: 41.5% of the Treasury roll is women.

End of session 1. Commit (`git add . && git commit -m "Pay gap: session 1, the six figures"`) and push. Tell Lars the one number that surprised you most.

Session 2 · the report

Exercise 5 By guild

15 min · `check5.py`

> **RAVEN #1205 from Orrin of the Clockworks:** *"One number for the whole realm hides everything. Where is the gap actually coming from? (If it's the Clockworks, I'd like to know before the Queen does.)"*

(a) Write `by_guild(df)`: loop over `df.groupby("department")` (the guild column); for each group collect `guild`, `headcount`, `pct_women` (1 decimal) and `median_gap` (with `gap_percent` on the medians) into a list of dicts.

(b) Turn the list into a DataFrame, `set_index("guild")`, sort by `median_gap` descending, return it.

(c) `check5.py` prints it.

EXPECTED OUTPUT

```
            headcount  pct_women  median_gap
guild
Kitchens           12       66.7        48.6
Clockworks         20       35.0        37.5
Harbour            25       24.0        36.0
Guard              35       40.0        31.1
Library            15       73.3        21.0
Forge              60       33.3        11.8
Treasury           18       50.0        11.0
Apothecary         15       53.3         2.6
```

Hint: The Kitchens are 67% women and have the biggest gap. That's not a contradiction; it's the whole point of the quartiles. Who are the four men in the Kitchens?

Exercise 6 The console report

20 min · `report_console.py`

> **RAVEN #1206 from Queen Aldara:** *"One sentence a queen can say out loud to the Privy Council without looking at a scroll. Then the detail underneath, for the councillors who like detail."*

(a) In `paygap.py`, write `one_sentence(df)`: pennies per crown is `round(100 - median_gap)`. Return exactly the sentence shown in the output below.

(b) Write `report_console.py`: load the data, call your five functions once each, and print the report below. The header line is `"=" * 50`; the percentages are right-aligned in 5 characters (`{value:>5}`).

EXPECTED OUTPUT

```
ROYAL TREASURY OF THORNMERE: ACCORD OF EQUAL COIN
==================================================
Subjects of the realm: 200 (83 women, 117 men)
Mean hourly pay gap:     19.7%
Median hourly pay gap:   18.7%
Mean bonus gap:          22.8%
Median bonus gap:        30.8%
Received a bonus:       women 45.8%, men 66.7%

gender        pct_women  pct_men
quartile
Lower              72.0     28.0
Lower middle       46.0     54.0
Upper middle       30.0     70.0
Upper              18.0     82.0

            headcount  pct_women  median_gap
guild
Kitchens           12       66.7        48.6
Clockworks         20       35.0        37.5
Harbour            25       24.0        36.0
Guard              35       40.0        31.1
Library            15       73.3        21.0
Forge              60       33.3        11.8
Treasury           18       50.0        11.0
Apothecary         15       53.3         2.6

Women of the realm earn 81 pennies for every crown earned by men (median hourly pay gap:
18.7%, mean: 19.7%).
```

Exercise 7 The chart and the workbook

25 min · `report.py`

> **RAVEN #1207 from Mistress Pell:** *"And now the version for the Council chamber: a picture of the quartiles and a workbook with the six figures on the first tab. Then we send it up to the throne room, and close the Grimoire."*

(a) Copy `report_console.py` to `report.py`. Add the matplotlib/openpyxl imports from lesson 11 and `OUT = Path("my_work/output")`, created with `mkdir`.

(b) Chart: a **stacked** bar per quartile. Women first (`plt.bar(quartiles.index, quartiles["pct_women"], label="Women")`), then men on top with `bottom=quartiles["pct_women"]`. Title, y-label, legend, save as `OUT / "quartiles.png"`.

(c) Workbook `OUT / "accord_of_equal_coin.xlsx"` with three sheets: **Summary** (a two-column DataFrame `measure` / `value` holding the six figures, `index=False`), **Quartiles**, **By guild**.

(d) Re-open it with openpyxl and, `for ws in wb.worksheets:`, make row 1 bold white on navy, set column A to width 28, freeze `A2`. Save. Print the last line below.

EXPECTED OUTPUT

Everything from Exercise 6, then:

```
Saved my_work/output/accord_of_equal_coin.xlsx and my_work/output/quartiles.png
```

Hint: Open the workbook and the PNG. That's the report. Have the Scriptorium copy the PNG, carry it up to the throne room, and watch the Queen's face.

Bug hunt Pages from the Grimoire, page 12

15 min · `bughunt/grimoire_bug.py`

The last page. Before he sailed west, Dalvric copied the Grimoire's pay gap incantation into Python. Every year since, it has told the Privy Council the gap is **24.6%**. Your report says 19.7%. The Queen wants to know who's right. **Three** bugs: one crashes, one divides by the wrong thing, one calls something a median that isn't.

EXPECTED OUTPUT ONCE FIXED

```
THE GRIMOIRE'S PAY GAP, CHECKED
Mean hourly gap:   19.7%
Median hourly gap: 18.7%
The Grimoire agrees with the Keeper. The Grimoire may now close.
```

One commit per bug. Then open the pull request, title: **"The Grimoire Closes"**. In the description, paste the Queen's sentence and the one thing you'd tell the Privy Council that the numbers don't say.

More riddles. Five more torn pages wait in `bughunt/riddles/`, one bug each (the last one has two), getting harder as you go. Optional: for fast finishers, or a quiet evening.